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RESEARCH ARTICLE

Intelligent Prediction of Flood Disaster Risk Levels Based on Knowledge Graph and Graph Neural Networks

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ABSTRACT Flash flood disasters pose a significant threat to human life and property, making accurate prediction of risk levels crucial for disaster prevention and mitigation. This study introduces an innovative artificial intelligence approach based on knowledge graphs and graph neural networks. The method integrates multi-source data to construct a knowledge graph, which is then modeled using graph neural networks. We evaluate the model's performance using metrics such as accuracy, precision, recall, F1 score, and AUC. Under five-fold cross-validation, the AUC reached 0.84, with all performance indicators showing good results, indicating significant performance improvement. Experimental results demonstrate high prediction accuracy when tested on a dataset containing 9000 records. Compared with the three classical models in traditional machine learning, such as RF, SVM and ANN, the performance of this model is improved, and it is better than the traditional model. Through case analysis, risk levels in multiple regions were accurately predicted. Additionally, statistical analysis of flood disaster warning levels and flash flood risk zoning across cities in Jiangxi Province provides a visual representation of flood risk distribution and risk level proportions in different cities, offering strong reference for flood prevention, disaster mitigation, and urban planning. This method provides important scientific support for precise flash flood disaster prediction and risk management.

INDEX TERMS Knowledge graph, graph neural network, flood disaster, risk level prediction.

I. INTRODUCTION

As a natural disaster with strong suddenness and great destructiveness, mountain torrent disasters pose a huge threat to people's lives and property safety [1]. According to statistics, the number of casualties and economic losses caused by mountain torrent disasters worldwide is huge every year [2]. In China, mountain torrent disasters occur frequently, especially in mountainous and hilly areas, which have a serious impact on the production and life of local residents [3]. The ''Outline of the 14th Five-Year Plan for National Economic and Social Development and Long-Range Objectives for 2035'' clearly states that it is necessary to strengthen the construction of natural disaster prevention and control capabilities and improve the capabilities of disaster prevention, mitigation, resistance, and relief [4]. Therefore, accurately predicting the risk level of mountain torrent disasters is of great significance for taking effective disaster prevention and mitigation measures in advance and reducing disaster losses.

Traditional methods for predicting the risk level of mountain torrent disasters mainly rely on empirical models and statistical analysis. Such methods often find it difficult to fully consider the complex factors and uncertainties in the formation process of mountain torrent disasters [5]. With the rapid development of artificial intelligence technology, especially the emergence of technologies such as knowledge graphs and graph neural networks, new ideas and methods are provided for predicting the risk level of mountain

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torrent disasters [6]. Knowledge graphs can represent entities, relationships, and attributes in the form of graphs, and can effectively integrate and express complex knowledge and information [7]. By constructing a knowledge graph related to mountain torrent disasters, multi-source data such as geographical location, topography, meteorological data, and historical disaster events can be correlated and fused to provide rich background knowledge and context information for risk level prediction [8], [9], [10], [11], [12], [13]. For example, in the process of promoting the construction of smart water conservancy, the country emphasizes the use of technologies such as big data and artificial intelligence to improve the level of water conservancy informatization. As an important knowledge representation and reasoning tool, knowledge graphs have broad application prospects in the field of water conservancy [14], [15], [16], [17], [18]. Graph neural networks can effectively process graph-structured data, capture the relationships and dependencies between nodes, and realize the modeling and analysis of complex systems [19], [20], [21], [22], [23], [24]. Applying graph neural networks to the prediction of mountain torrent disaster risk levels can fully utilize the information in knowledge graphs, dig out potential risk patterns and laws, and improve the accuracy and reliability of predictions [25], [26]. In recent years, the country has increased its support for scientific and technological innovation, encouraged scientific researchers to carry out in-depth research in fields such as disaster prediction, and promoted the development and application of related technologies [27], [28], [29], [30], [31], [32].

In conclusion, this study aims to combine the advantages of knowledge graphs and graph neural networks to construct an artificial intelligence-based mountain torrent disaster risk level prediction model to improve the accuracy and efficiency of predictions and provide scientific basis and decision support for disaster prevention and mitigation work.

II. RELATED WORK

A. RESEARCH STATUS OF MOUNTAIN TORRENT DISASTER RISK ASSESSMENT METHODS

Mountain torrent disaster risk assessment is an important foundation for disaster prevention and mitigation. In recent years, many scholars have been committed to researching mountain torrent disaster risk assessment methods. Traditional methods mainly include methods based on hydrological models [33], [34], [35], [36], methods based on empirical formulas [37], and methods based on index systems [38]. These methods can assess the risk of mountain torrent disasters to a certain extent, but there are problems such as large data requirements, complex calculations, and difficulty in considering the interaction of multiple factors. With the development of information technology, some new methods such as methods based on Geographic Information System (GIS), methods based on remote sensing technology [39], and methods based on numerical simulation have been widely used. These methods can obtain and analyze data more accurately and improve the accuracy of risk assessment, but there are still some limitations, such as high requirements for data quality and difficulty in determining model parameters.

In addition, some studies use machine learning and data mining methods for mountain torrent disaster risk assessment. For example, support vector machine (SVM), random forest (RF), artificial neural network (ANN) and other methods are applied in the risk assessment of mountain floods [40]. Among them, Wu Xiaojun, Fang Xiuqin and other took Jiangxi Province as an example, selected 9 influencing factors affecting mountain flood disaster and used random forest algorithm to assess the disaster risk of mountain flood in Jiangxi Province. The results showed that the results of mountain flood disaster risk assessment could meet the requirements; Ma Xuting constructed risk identification model based on Support Vector Machine (SVM) and random forest, selected RF as mountain flood disaster evaluation model, and divided mountain flood disaster types into 4 categories by considering local terrain and confluence [41]. These methods are able to automatically learn features and patterns from data, improving the accuracy of predictions, but often difficult to handle complex nonlinear relationships and multimodal data. Liu Fang trained the artificial neural network by using the typhoon events, and applied it to the typhoon and rainstorm prediction in Yongjia County, Zhejiang Province, to realize the dynamic prediction of the flood disaster caused by the typhoon and rainstorm.

B. APPLICATIONS OF KNOWLEDGE GRAPHS AND GRAPH NEURAL NETWORKS IN RELATED FIELDS

Knowledge graphs show strong application potential in multiple fields, such as intelligent question answering, recommendation systems, and medical diagnosis. In the field of disaster management, knowledge graphs can integrate various disaster-related information and provide support for disaster prediction and response [42]. For example, some studies use knowledge graphs to construct semantic networks of disaster events to better understand the occurrence mechanism and influencing factors of disasters [27]. Graph neural networks have achieved remarkable results in fields such as image recognition, natural language processing, and social network analysis. In terms of disaster prediction, graph neural networks can process complex relationship data, mine potential risk patterns, and provide new ideas for disaster prediction. For example, some studies have applied graph neural networks to the prediction of earthquake disasters and improved the accuracy of predictions by analyzing the correlation between earthquake events. In terms of mountain torrent disaster risk level prediction, there are also some related research works. For example, some scholars use traditional machine learning methods combined with geographic information data to predict the occurrence risk of mountain torrent disasters [43]. There are also some studies that try to apply deep learning methods to the prediction of mountain torrent disasters, but most of these methods do not fully

consider the interaction of multiple factors and complex relationships in the formation process of mountain torrent disasters [44].

In conclusion, although existing methods have achieved certain results in mountain torrent disaster risk assessment, further improvement and innovation are still needed. This study introduces knowledge graphs and graph neural networks into mountain torrent disaster risk level prediction, aiming to fully utilize multi-source data, mine potential risk patterns and laws, and improve the accuracy and reliability of predictions.

This paper collects the information including the investigation and evaluation data of mountain flood disasters, the detailed description of typical mountain flood disaster events, and related news reports. The specific data sources are provided as follows:

(1) Mountain flood disaster survey and evaluation data of Jiangxi Province, a total of 9758 mountain flood disaster hazard area data, including geographical location, hazard area population, current flood control capacity, risk type, etc., 9000 items were finally obtained after excluding noise;

(2) In the past 15 years, there were 1,756 news reports with the title of mountain torrent disaster keywords, and 857 articles were finally eliminated from noise;

(3) Jiangxi Province 2022 Annual Statistical Yearbook, Statistical Bulletin, etc.

III. MODEL CONSTRUCTION

A. METHOD OVERVIEW

This paper proposes a risk level prediction method based on knowledge graph and graph neural network, abbreviated as GNN-Risk. This method includes four modules: data preprocessing, knowledge graph construction, graph neural network modeling, and risk level prediction, as shown in Figure 1. By combining knowledge graph and graph neural network, it can improve the accuracy of prediction while handling complex relationships and structured data.

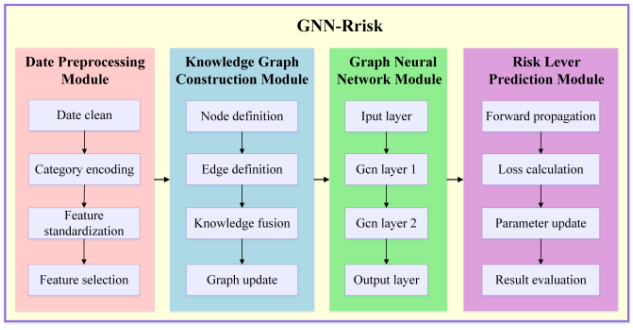


FIGURE 1. Architecture diagram of the GNN-Risk model.

B. DATE PREPROCESSING MODULE

In risk prediction tasks, the quality of data is crucial to the performance of the model. Therefore, the data preprocessing module is the fundamental link in the entire model construction process. It mainly includes four steps: data cleaning, category encoding, feature standardization, and feature selection. First, in the data cleaning stage, we remove missing values and outliers in the data to ensure the integrity and consistency of the data, thus ensuring that the input data of the model is of high quality. Next, for the categorical features in the data, category encoding converts these categorical variables into numerical features that can be used for model training. This step is very crucial for subsequent feature input. In the feature standardization part, numerical features are standardized to conform to the standard normal distribution to reduce the scale differences between different features and avoid the impact on the model due to different ranges of feature values.The data standardization is calculated according to Formula (1). Finally, through feature selection, we select the features that have a greater contribution to model training, thereby reducing the feature dimension and optimizing the model performance.

$$x' = \frac{x - u}{\sigma} \tag{1}$$

where x' Representing standardized features, X representing original features, μ and σ represent the mean and standard deviation of the features respectively.

Through this series of data preprocessing operations, we have laid a solid foundation for the construction of subsequent modules. Especially in the knowledge graph construction module, these cleaned, encoded, and selected features will be used as the feature input of knowledge graph nodes.

C. KNOWLEDGE GRAPH CONSTRUCTION MODULE

After the data preprocessing is completed, the processed feature information is further used for the construction of a knowledge graph. The knowledge graph construction module performs structured modeling on these data and specifically includes four steps: node definition, edge definition, knowledge fusion, and graph update. First, each entity in the data (such as geographical location, risk type, etc.) is defined as a node in the graph, and the basic structure of the graph is clarified through node definition. On this basis, the edges between nodes, that is, the relationships between entities, are defined, such as the association between geographical location and its corresponding risk type. After that, through the knowledge fusion step, external data (such as historical flood data and geographical information, etc.) is integrated into the knowledge graph to further enrich the content of the graph. Finally, with the continuous influx of new data, the graph structure is dynamically updated through the graph update mechanism to ensure its timeliness and accuracy.The construction formula of the knowledge graph is as follows:

$$G = (V, E) \tag{2}$$

where G represents the knowledge graph, V represents the node set, and E represents the edge set.

The calculation formula for edge weight is:

$$w_{ij} = sim(v_i, v_j) \tag{3}$$

where w_{ij} represents the edge weight between node v_i and v_j, and $sim(v_i, v_j)$ represents the similarity between nodes.

Through the construction of the knowledge graph, the complex association relationships in the data are modeled in the form of a graph structure, providing structured input for the next graph neural network module.

D. GRAPH NEURAL NETWORK MODULE

After the construction of the knowledge graph is completed, the graph neural network module is responsible for learning and aggregating the node features in the graph and further extracting feature information for risk level prediction. This module consists of an input layer, two layers of GCN (Graph Convolutional Network) layers, and an output layer. First, in the input layer, the node features and adjacency matrix are taken as the input of the model to ensure that the graph neural network can capture the structured relationships between nodes. Then, the node features are aggregated and updated layer by layer through two layers of GCN. Each layer of GCN uses the adjacency matrix and degree matrix to aggregate the information of neighboring nodes and update the feature representation of nodes. The formula is as follows:

$$H^{(l+1)} = \sigma(D^{-1/2}AD^{-1/2}H^{(l)}W^{(l)}) \tag{4}$$

where $H^{(1)}$ represents the node feature matrix of the l-th layer, A represents the adjacency matrix, D represents the degree matrix, $W^{(l)}$ represents the weight matrix of the l-th layer, and σ represents the activation function.

Node feature update formula:

$$h_i = \sum_{j \in N(i)} \frac{h_j}{\sqrt{d_i d_j}} \tag{5}$$

where h_i' represents the updated feature of node i, N (i) represents the set of neighboring nodes of node i, and d_i and d_j respectively represent the degrees of nodes i and j.

In this way, the graph neural network can effectively learn the local features of nodes and map these updated features to specific risk levels in the output layer.

E. RISK LEVEL PREDICTION MODULE

After the graph neural network completes the learning and update of node features, the risk level prediction module is responsible for predicting the risk level and evaluating the performance of the model. This module includes four steps: forward propagation, loss calculation, parameter update, and result evaluation. First, through forward propagation, the node features are input into the GCN layer, and after multiple layers of calculation, the prediction result of the risk level is obtained. Then, the cross-entropy loss function is used to calculate the error between the prediction result and the true label. The formula of the cross-entropy loss function is as follows:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{c} y_{ic}\log(\hat{y}_{ic}) \tag{6}$$

where N represents the number of samples, C represents the number of categories, y_{ic} represents the true label of sample i, and $\hat{y}_{ic}$ represents the predicted probability of sample i.

Then, the model parameters are updated through the gradient descent algorithm to minimize the loss function. The formula is as follows:

$$\theta = \theta - \eta\nabla_\theta L \tag{7}$$

where θ represents the model parameters, η represents the learning rate, and ∇_θL represents the gradient of the loss function with respect to the parameters.

IV. EXPERIMENTS

A. OVERVIEW OF THE STUDY AREA AND DATA SOURCES

Jiangxi Province has a subtropical monsoon climate with dense rivers and streams. The annual average precipitation is 1638 mm. The unique climatic characteristics and topographic and geomorphic conditions lead to frequent, recurring, and severe mountain torrent disasters. According to the statistics of the provincial mountain torrent disaster survey and evaluation data, the area of mountain torrent disaster prevention zones in the province reaches 13.02×10^4 km^2, accounting for 78% of the total area of the province. There are 9758 mountain torrent disaster risk areas, and the population in the risk areas is more than 800,000. The data set in this paper has a wide range of sources, including mountain torrent disaster survey and evaluation data, news reports, statistical yearbooks, etc. Among them, the Jiangxi Province mountain torrent disaster survey and evaluation data totals 9758 items. After removing noise, 9000 items are obtained; there are 1756 news reports with mountain torrent disasters as keywords in the past 15 years. After removing noise, 857 items are obtained. The experimental data set contains 9000 records with features such as geographical location and risk type. It is divided into a training set of about 7200 items, a validation set of 900 items, and a test set of 900 items in a ratio of approximately 8:1:1, which are used for model training, model selection and hyperparameter tuning, and evaluating the generalization performance of the model respectively.

B. KNOWLEDGE GRAPH CONSTRUCTION

In the process of knowledge graph construction, we define geographical location, risk type, historical events, etc. as nodes and define the relationships between them. The finally constructed knowledge graph contains 2000 nodes and 3000 edges. The information in the knowledge graph provides rich context information for the graph neural network and helps improve the prediction performance of the model. The knowledge graph of mountain flood prevention and control information is shown in Figure 2.

In this diagram, not only the nodes of each hazard zone are shown, but also the complex relationships between these

nodes are presented through the connection of edges, such as the association with other nodes such as geographical location, risk type, flood control capacity, population number, etc. Through the display of this complex relationship, it can not only visually reflect the potential transmission paths of disaster risks between various regions, but also more comprehensively show the interaction between the various data types involved in the study. The interconnection of nodes and edges in the map provides deeper visualization support for understanding the spatial distribution and influencing factors of flash flood disaster risk. This presentation helps to reveal the potential correlations between the data, providing a key tool for further analysis.

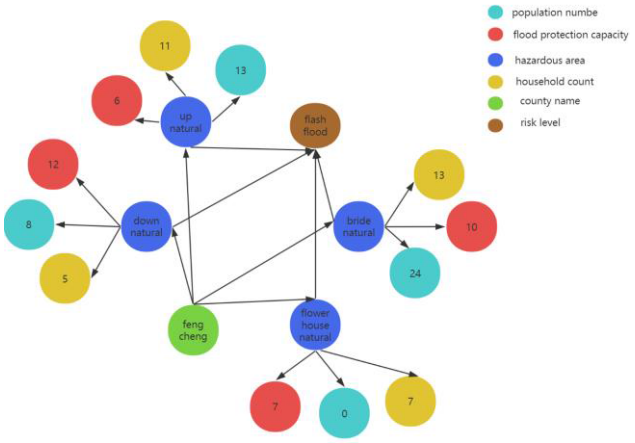


FIGURE 2. Complex knowledge graph for flash flood disaster prediction.

C. EXPERIMENTAL SETUP

In the experiment, two layers of GCN are used, the number of hidden units in each layer is 64, and the activation function is ReLU. The Adam optimizer is used for model training, the learning rate is 0.01, and the number of training epochs is 1000. During the training process, an early stopping strategy is adopted to prevent overfitting. The training environment is Python 3.8, and the implementation is carried out using the PyTorch framework. As shown in the figure is the loss curve in the experimental training process. It can be seen that as the number of training epochs increases, the model gradually converges. The model training loss curve is shown in Figure 3.

In the later stage of model training (after approximately 200 rounds), the loss curve starts to exhibit slight oscillations. As the number of training rounds increases, the amplitude of the oscillations gradually decreases while the frequency of the oscillations gradually speeds up. Eventually, the loss value tends to converge. This phenomenon indicates that during the training process, the model gradually adapts to the complexity of the data through multiple adjustments. The gradual weakening of these oscillations shows the stability of the model when finely learning complex features. The reduction

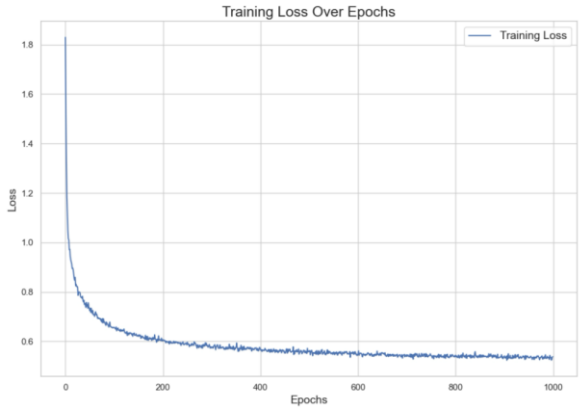


FIGURE 3. Model training loss curve.

in the oscillation amplitude reflects the gradual convergence of model parameter adjustments. Eventually, it tends to be stable, indicating that the model has reached an optimal state.

D. EXPERIMENTAL DEMONSTRATION

When evaluating model performance, in addition to using traditional classification indicators such as accuracy, precision, recall, and F1-Score, the area under the receiver operating characteristic curve (AUC) is also used to further measure the performance of the model, the calculation formula is as follows. AUC is an important indicator for measuring the sorting ability of a classification model for positive and negative samples. Its value range is between 0.5 and 1. A value of 1 indicates that the model has perfect discrimination ability, while a value of 0.5 indicates that the model has no discrimination ability, which is equivalent to random guessing. Specifically, the closer the AUC is to 1, the stronger the classification ability of the model. In this study, we performed 5-fold cross-validation on the model to evaluate its performance more robustly. The AUC value of each fold is shown in the table and gradually increases, rising from 0.79 in the first fold to 0.84 in the fifth fold. At the same time, other performance indicators such as accuracy, precision, recall, and F1-Score also show corresponding improvements. These results indicate that with the training of the model, the overall performance of the model has been significantly improved, verifying the effectiveness of the method we proposed. The model prediction results are shown in Table 1.

1.Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

2.Precision:

$$\Pr ecision = \frac{TP}{TP + FP} \tag{9}$$

3.Recall:

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

4.F1 Score:

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{11}$$

In equations 8–11, TP represents the number of samples that the model correctly predicted as a positive example. TN represents the number of samples that the model has correctly predicted as negative cases. FP represents the number of samples that the model mispredicted as a positive example. TN represents the number of samples where the model mispredicted a negative case.

TABLE 1. Model prediction results.

***	AUC	Accuracy	Precision	Recall	F1-Score
Fold 1	0.79	0.85	0.83	0.84	0.83
Fold 2	0.81	0.86	0.84	0.85	0.84
Fold 3	0.82	0.87	0.85	0.86	0.85
Fold 4	0.83	0.88	0.86	0.87	0.86
Fold 5	0.84	0.89	0.87	0.88	0.87

In the experiment, the 5-fold cross-validation method is used to evaluate the model. Specifically, the data set is randomly divided into 5 subsets. Each time, 4 subsets are used for training and 1 subset is used for validation. Repeat 5 times, and the final result is the average of the 5 experiments. Cross-validation effectively evaluates the stability and generalization ability of the model. The Flash flood disaster risk assessment model 5 fold cross-verified ROC curve is shown in Figure 4.

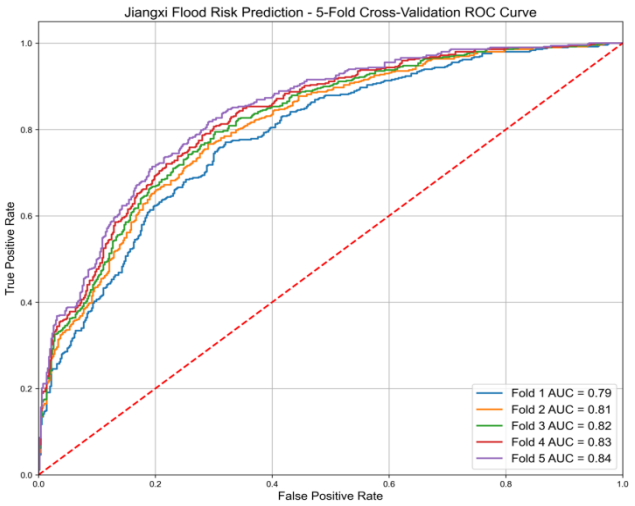


FIGURE 4. Flash flood disaster risk assessment model 5 fold cross-verified ROC curve.

E. MODEL COMPARISON

In order to prove the effectiveness of the GNN-Risk model based on knowledge graph and graph neural network proposed in this paper, the algorithm is compared with three classical models of RF, SVM and ANN in traditional machine learning, and the specific output results are shown in Table 2.

TABLE 2. Model contrast results.

Model	AUC	Accuracy	Precision	Recall	F1-Score
GNN-Risk	0.84	0.89	0.87	0.88	0.87
RF	0.76	0.79	0.80	0.82	0.84
SVM	0.78	0.81	0.83	0.85	0.82
ANN	0.81	0.86	0.81	0.82	0.84

Table 2 shows that these models all have high accuracy in risk prediction, but the accuracy, precision, recall and F1 scores are lower than the GNN-Risk model presented here. In conclusion, the GNN-Risk model proposed in this paper has better performance compared with the previous machine learning model and can be used for flood disaster.

F. ABLATION EXPERIMENT

The ablation experiment aims to explore the contribution of each module and feature to the overall performance of the model, and to evaluate the impact on the evaluation effect by gradually removing or replacing the key components of the model, so as to verify the necessity and effectiveness of each part. The following four sets of ablation experiments were designed in this study:

1. Remove the knowledge graph module;
2. Replace the graph neural network with a traditional machine learning model such as RF;
3. Remove the relationship between the nodes, and train only with the node features;
4. Remove the data preprocessing step;

The specific ablation experimental operation procedures and results are shown in Table 3.

TABLE 3. Results of ablation experiments.

experiment number	experiment content				AUC
	Step1	Step2.1	Step2	Step3	
1	√	√		√	0.78
2	√	√	√		0.76
3	√		√	√	0.80
4		√	√	√	0.74
5	√	√	√	√	0.84

According to the results of ablation experiments, it can be found that the knowledge graph and graph neural network modules are the key parts to improve the performance of the model, among which the knowledge graph module is particularly effective on integrating data correlation and background information, while the graph neural network can effectively capture the complex relationships and dependencies between nodes. This indicates significant model architecture based on knowledge graph and graph neural network in mountain torrent disaster risk assessment.

V. CASE STUDY

A. VISUALIZATION OF FLOOD DISASTER RISK LEVEL PREDICTION IN JIANGXI PROVINCE

Figure 5 shows the distribution of flood disaster warning levels in various cities in Jiangxi Province, and conducts

a comprehensive analysis and visual display of the flood warning levels of various cities in the province. In the figure, a gradient color system from green to red is used to represent different flood risk levels. The colors from low to high represent flood risk levels of lower (risk level 3), medium (risk level 2), and higher (risk level 1) in turn. It can be clearly seen from the figure that the high-risk areas in Jingdezhen City and Shangrao City are more prominent. 70% and 65% of the areas are at a higher risk level (risk level 1), indicating that these areas are facing a greater risk of mountain flood disasters. In addition, 70% and 58% of the areas in Ganzhou City and Jiujiang City are at a medium risk level (risk level 2), indicating that the flood risk in these cities is also at a medium level, and prevention work needs to be strengthened. Nanchang City is dominated by low-risk areas. About 72% of the areas are at a lower risk level (risk level 3), indicating that its flood risk is relatively low. For other cities such as Pingxiang and Fuzhou, although the risk level distribution is relatively uniform, they still need to be vigilant about high-risk areas to ensure that flood control measures can be in place in time.

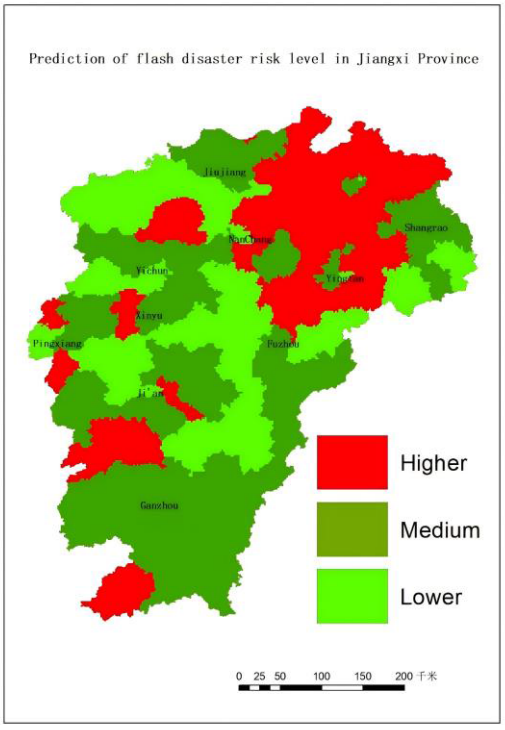


FIGURE 5. Prediction of mountain flood disaster risk levels in Jiangxi Province.

B. STATISTICAL ZONING OF MOUNTAIN FLOOD RISK LEVEL

Figure 6 shows the percentage distribution of each city in Jiangxi Province under three mountain flood disaster risk levels. As can be seen from the stacked column chart, the risk distribution of Nanchang City is mainly at a lower risk level (risk level 3), and about 72% of the area belongs to a lower risk level, indicating that the possibility of mountain flood disasters in Nanchang City is relatively low. In contrast, the risk distribution of Jingdezhen City and Shangrao City is mainly at a higher risk level (risk level 1), and 70% and 65% of the areas respectively belong to a higher risk level, showing that these areas are facing a greater threat of mountain flood disasters. The risks of Ganzhou, Jiujiang and Yichun cities are concentrated at the medium risk level (risk level 2). 70%, 58% and 50% of the areas in Ganzhou, Jiujiang and Yichun respectively are at the medium risk level, indicating that these cities need to strengthen prevention measures for mountain flood disasters. The risk distribution of other cities is relatively uniform. For example, 40% of the area in Pingxiang City is at a higher risk, while 55% of the area in Xinyu City belongs to a lower risk level. In general, the risk level distributions of different cities are significantly different. Nanchang City is mainly at a lower risk, while the proportion of higher risks in Jingdezhen and Shangrao cities is significantly higher than that in other cities, providing an important reference for the planning and disaster prevention and control of these cities.

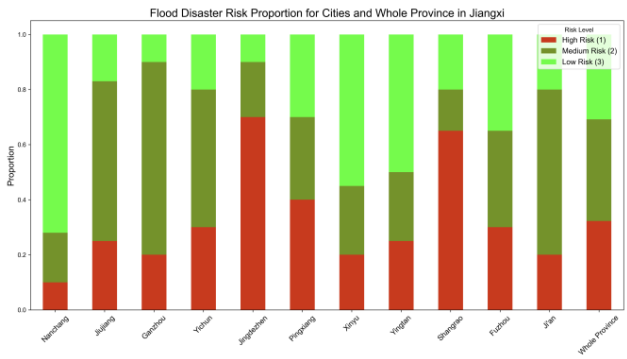


FIGURE 6. Percentage of area occupied by different risk levels in each prefecture-level city.

VI. CONCLUSION

This paper proposes a mountain flood disaster risk level prediction method based on knowledge graph and graph neural network. This method significantly improves the efficiency of data integration and complex relationship analysis. Especially when facing multi-source heterogeneous data, it can effectively solve the problem of complex associations between data. By combining historical mountain flood disaster data with the constructed risk assessment model, the accuracy of this model reaches 84%. The algorithm is compared with three classical models in traditional machine learning, such as RF, SVM and ANN, and the model has been improved, indicating its high reliability in predicting the mountain flood disaster risk level in Jiangxi Province. The successful application of this method not only provides strong support for the accurate prediction of mountain flood disaster risks, but also lays a solid foundation for further risk assessment and disaster prevention decision-making.

From the analysis results of the distribution of risk areas, the proportions of medium-risk areas in Ganzhou City and Jiujiang City reach 70% and 58% respectively, indicating that the probability of mountain flood disasters in these two cities is relatively high and the potential losses are relatively serious. Based on this, it is recommended to strengthen flood control measures in advance and make emergency preparations in these areas to deal with possible floods. In contrast, the proportions of higher-risk areas in Jingdezhen City and Shangrao City reach 70% and 65% respectively, showing that the risk of mountain flood disasters in these cities is more significant. To this end, it is recommended that relevant departments deploy disaster prevention and mitigation measures as early as possible in these areas to reduce the losses and risks caused by floods. The risk level of Nanchang City is relatively low. About 72% of the area is at a lower risk level, indicating that the possibility of mountain floods in this city is relatively small. Nevertheless, monitoring and early warning work still needs to be strengthened in low-risk areas.

Through in-depth analysis of the distribution of mountain flood disaster risk levels in various cities in Jiangxi Province, this paper provides valuable data support for the planning and emergency management work of each city. Especially for cities in high-incidence areas of mountain flood disasters, this paper proposes targeted disaster prevention strategies and suggestions. This risk prediction method provides a solid foundation for improving the scientific and effectiveness of disaster management. In the future, by further strengthening dynamic monitoring and timely early warning of high-risk areas, the accuracy and foresight of disaster prevention and reduction work can be continuously improved to ensure that each city can respond quickly and effectively prevent and control potential disasters.

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